CSE310: Compiler Sessional, January 2026 Term
ICG Online Assignment Practice
Ternary Conditional Operator

Time: 40 Mins **Total Marks: 10**

Problem

Extend the grammar to support the C-style ternary conditional operator (`? :`). Your program should evaluate the condition and short-circuit the execution to only evaluate the correct expression branch. Thus, your program should support the following syntax.

```
int x, y; x = 10; y = (x > 5) ? 100 : 200;
```

Task

- Extend the current grammar to support the ternary operator in expressions.
- Update your ICG code to generate jumping assembly code for this feature. If the condition is true, jump to the first expression. Otherwise, jump to the second. Both expressions will leave their result in `EAX`.

Mark Distribution

- Extending the grammar to support the ternary conditional operator. (4)
- Implementing conditional jumping and code generation logic. (6)

Sample Input-Output

Input Program	Generated Sample Assembly (print library omitted)

<table>
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```
int main() {
    int x, y;
    x = 10;
    y = (x > 5) ? 100 : 200;
    println(y);
    return 0;
}
```

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```
main:
    PUSH EBP
    MOV EBP, ESP
    SUB ESP, 8
    MOV EAX, 10
    MOV [EBP-4], EAX
    MOV EAX, [EBP-4]
    MOV EDX, EAX
    MOV EAX, 5
    CMP EDX, EAX
    JG .L1
    JMP .L2
.L1:
    MOV EAX, 100
    JMP .L3
.L2:
    MOV EAX, 200
.L3:
    MOV [EBP-8], EAX
    MOV EAX, [EBP-8]
    CALL print_number
    MOV EAX, 0
    JMP .L4
.L4:
    ADD ESP, 8
    POP EBP
    RET
```

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<tr><td>Expected Output</td><td>

```
100
```

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